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Activation of N-ras and K-ras induced by interleukin-6 in a myeloma cell line: implications for disease progression and therapeutic response

Matt Rowley[1] and Brian Van Ness*,[2]

[1]*Graduate Program in Molecular, Cellular, Developmental Biology and Genetics, The University of Minnesota, Minneapolis, Minnesota, MN 55455, USA;* [2]*The Department of Genetics, Cell Biology and Development, The University of Minnesota, Minneapolis, Minnesota, MN 55455, USA*

Ras is a major signaling molecule activated by interleukin-6. There have been no published reports, however, that specifically examine the kinetics and percentage of Ras activation in response to IL-6. Model cell lines were used to study activation of N- and K-ras induced by IL-6. All of the myeloma cell lines we tested express both N-ras and K-ras, but not H-ras. GTP-bound Ras was measured and the percentage of the total Ras pool that was activated in response to IL-6 was calculated. IL-6 is able to transiently activate both N- and K-ras in the ANBL6 cell line. In addition, increasing concentrations of IL-6 are able to activate increasing levels of both N- and K-ras. One ng/ml of IL-6 is able to activate approximately 10% of the N-ras pool and 18% of the K-ras pool. The amount of Ras-GTP in the cells correlates with the level of proliferation at low levels, but proliferation plateaus when higher levels of Ras-GTP are present. Protection from dexamethasone-induced apoptosis correlates with IL-6 concentration and Ras activation. However, IL-6 enhances apoptosis induced by doxorubicin. Interestingly, the ANBL6 cell line transfected with an N-*ras12* or a K-*ras12* gene is protected from doxorubicin-induced apoptosis.
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Introduction

Signals transmitted from the cell surface to the nucleus are essential components of cell viability. A complex network of signal transduction pathways link cell surface receptors to the control of cellular proliferation, differentiation, and survival. The Ras family, comprised of H-*ras*, N-*ras* and K-*ras*, are critical components of many of these signal transduction pathways (Hall, 1998; McCormick, 1996).

H-, N-, and K-ras are small monomeric G proteins that act as molecular switches (Takai *et al*., 2001). Ras cycles between the GDP-bound inactive state and the GTP-bound active state. An upstream signal from an activated receptor leads to the dissociation of GDP from the inactive Ras and subsequent association of GTP. This induces a conformational change in the effector-binding domain of Ras and leads to activation of downstream effectors. The intrinsic GTPase activity of Ras converts the GTP-bound Ras back to the inactive GDP-bound form completing one cycle of Ras activation. Point mutations at codons 12, 13, or 61 lead to a constitutively active, GTP-bound Ras protein.

Interleukin-6 (IL-6) has been shown to act as a growth factor for multiple myeloma cells (Hirano *et al*., 1997; Kawano *et al*., 1998; Klein *et al*., 1989). In addition, IL-6 has been shown to protect myeloma cells from apoptosis induced by dexamethasone, anti-Fas antibodies, and serum starvation (Chauhan *et al*., 1997a; Hardin *et al*., 1994; Lichtenstein *et al*., 1995). IL-6 signals through a number of different downstream effectors including Ras (Nakafuku *et al*., 1992). Several reports have demonstrated the presence of activating mutations in N- and K-ras, but not H-ras, in myeloma cells with a frequency as high as 40–50% (Corradini *et al*., 1993; Liu *et al*., 1996; Neri *et al*., 1989; Paquette *et al*., 1990; Portier *et al*., 1992). It has been shown that activating mutations in either N- or K-*ras* can provide a proliferation signal independent of any additional IL-6 in the ANBL6 myeloma cell line (Billadeau *et al*., 1995, 1997). In addition, *ras* mutations have been shown to protect myeloma cell lines from apoptosis induced by dexamethasone, doxorubicin, or melphalan (Rowley *et al*., 2000). This demonstrates the importance of Ras dependent signal transduction in myeloma cells.

Even though IL-6-induced signal transduction through Ras has been well documented, there have not been any published reports of specific kinetics of Ras expression and activation in myeloma cells. In this report we examined the expression of H-, N-, and K-ras in several myeloma cell lines. In addition, we studied the kinetics of Ras activation induced by IL-6 over time, or by increasing concentrations of IL-6. We were able to calculate the percentage of the total cellular N- or K-ras pool that was activated in

*Correspondence: Dr B Van Ness, Cancer Center Research Building, Mayo Mail Code 806, Minneapolis, Minnesota, MN 55455, USA; E-mail: Vanne001@tc.umn.edu
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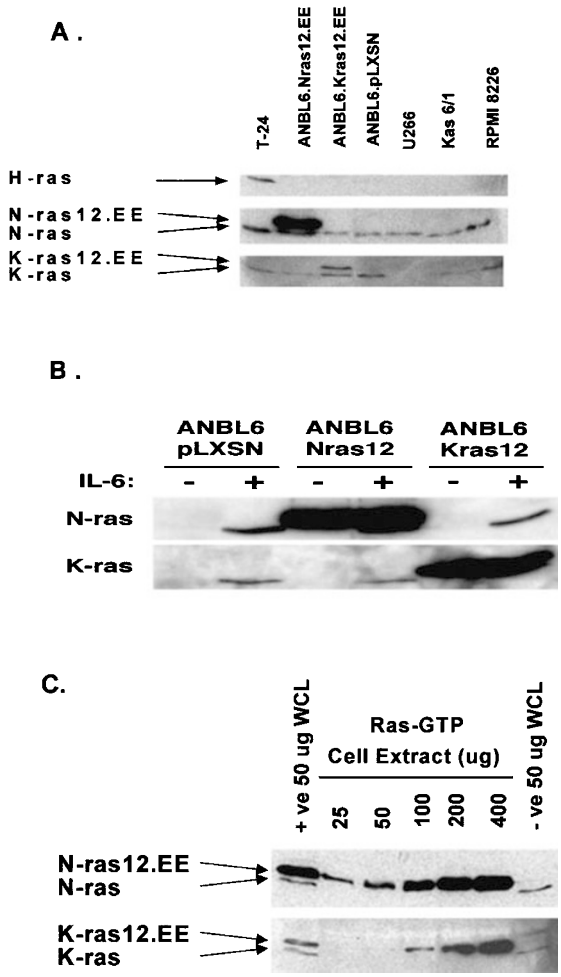

Figure 1 Ras expression and IL-6 induced activation in myeloma cell lines. (**a**) Myeloma cell lines express N-ras and K-ras, but not H-ras. Whole cell extracts were prepared from ANBL6.pLXSN, U266, Kas 6/1 and RPMI 8226 myeloma cell lines. Ras expression was detected by Western blot using antibodies specific for H-ras, N-ras, or K-ras. The T-24 cell line was used as a positive control for H-ras expression. ANBL6 cell lines transfected with the N-ras12.EE gene or the K-ras12.EE gene were used as controls for N-ras and K-ras expression respectively. Two bands appear with these cell lines because the transfected Ras has an epitope tag causing it to migrate slower through the gel than the endogenous Ras. (**b**) IL-6 induces the activation of both N-ras and K-ras. ANBL6.pLXSN, ANBL6.Nras12.EE or ANBL6.Kras12.EE cells were stimulated with or without IL-6 for 15 min. Cell extracts were prepared and Ras activation was measured using ARIA. There are undetectable levels of activated N- and K-ras in the ANBL6.pLXSN cell line in the absence of IL-6. Addition of IL-6 leads to the activation of both N- and K-ras. The ANBL6.Nras12.EE and ANBL6.Kras12.EE cell lines showed constitutive N- or K-ras activation respectively even in the absence of IL-6 due to the activating mutation at codon 12. Addition of IL-6 resulted in the activation of endogenous N- and K-ras as measured by the faster migrating band. (**c**) The ARIA assay is semi-quantitative. Extracts were prepared from ANBL6.Nras12.EE or ANBL6.kras12.EE cells. The protein concentration was measured and various amounts of protein were used to measure activated Ras by ARIA. 50 μg of whole cell extracts from ANBL6.Nras12.EE (top) or ANBL6.Kras12.EE (bottom) were used as a positive control for expression of N-ras12 or K-ras12. The efficiency of the ARIA assay was measured by comparing the density of the Ras-GTP band to the expected density based on total N-ras12 or K-ras12 expression

response to IL-6 and correlate this with proliferation and therapeutic response.

Results

Principle of the activated Ras interaction assay

Activation of Ras is induced by a number of different stimuli including cytokines, growth factors, and adhesion molecules. Assays that have been used to study Ras activation include thin-layer chromatography (Brownell *et al*., 1997; Hata *et al*., 1993; Jones and Jackson, 1998) and the activated Ras interaction assay (ARIA) (Taylor and Shalloway, 1996). In the ARIA assay, the Ras binding domain of Raf-1 is fused to a GST protein and bound to glutathione beads. This complex will only bind to GTP-bound Ras and can be used to precipitate Ras-GTP from cell lysates. Affinity-precipitated Ras is then detected by Western blotting with antibodies specific to H-, N-, or K-ras.

Interleukin-6 is able to activate both N- and K-ras

IL-6 has been shown to activate Ras in myeloma cell lines; however, there have not been any published reports that have distinguished between activation of H-ras, N-ras, or K-ras. In order to determine which Ras proteins are activated by IL-6 we first needed to determine which forms of Ras are expressed in our cell lines. Whole cell extracts were made from the ANBL6, U266, Kas 6/1 and RPMI 8226 myeloma cell lines. In addition, extracts were made from T-24 (bladder cancer cell line containing the H-ras oncogene) and the ANBL6 cell line transfected with either an N-ras12.EE or K-ras12.EE gene. Western blot analysis shows that N- and K-ras are expressed in the four myeloma cell lines tested, but that H-ras is not (Figure 1a). Overexposure of the K-ras blot shows that K-ras is weakly expressed in the U266 cell line. The ANBL6.Nras12.EE and ANBL6.Kras12.EE cell lines express either an epitope tagged N-ras or K-ras respectively. Since the epitope tag slows the migration through the gel, these proteins can be distinguished from endogenous N- or K-ras and act as controls for antibody specificity. Data from four independent experiments show that protein expression of the N-ras12.EE is about 3.3-fold higher than the endogenous N-ras and expression of K-ras12.EE is about 1.3-fold higher than endogenous K-ras.

Since both N- and K-ras are expressed in the ANBL6 cell line, we wanted to determine whether IL-6 was able to activate both N- and K-ras. Figure 1b shows that in the absence of IL-6, there is no detectable GTP-bound N- or K-ras in the ANBL6.pLXSN cell line. Addition of IL-6 induces activation of both N- and K-ras. The ANBL6.Nras12.EE and ANBL6.Kras12.EE cell lines show constitutive activation of N- or K-ras respectively. In addition, stimulation with IL-6 induced activation of

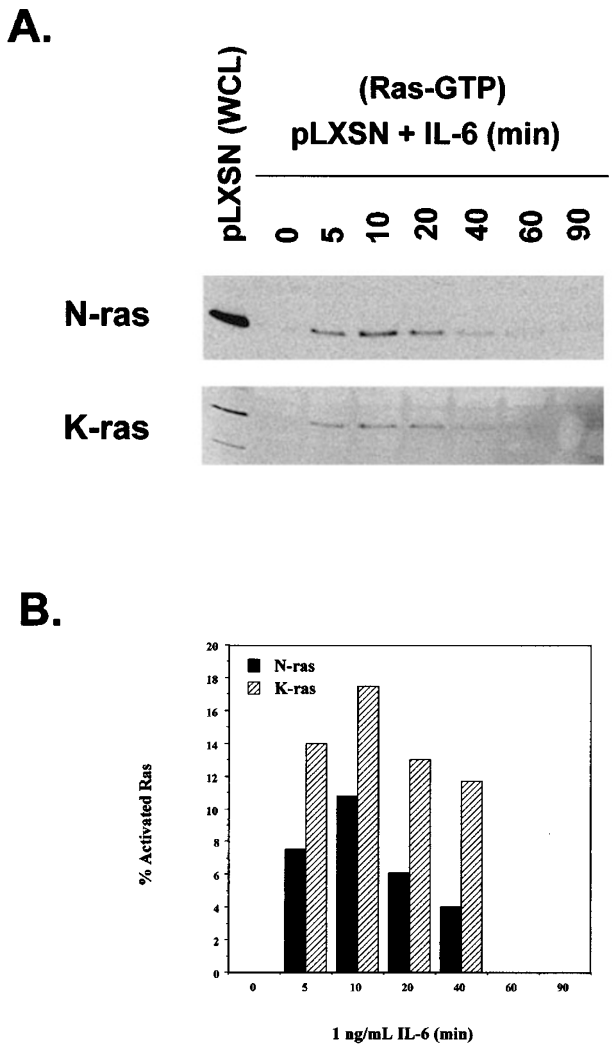

Figure 2 Interleukin-6 induces transient activation of N-ras and K-ras. (**a**) ANBL6.pLXSN cells were stimulated with 1 μg/ml IL-6. Cells were harvested at the indicated time point and extracts were made to measure Ras activation by ARIA using 400 μg of extract. 40 μg of whole cell lysates were used in lane 1 to measure total Ras expression. (**b**) The percentage of activated ras was measured by comparing the density of the N- or K-ras-GTP band with the total N- or K-ras expression. The percentage was adjusted to account for the 10-fold higher protein concentration used for ARIA as well as the 35 or 39% efficiency of ARIA for N- or K-ras activation respectively

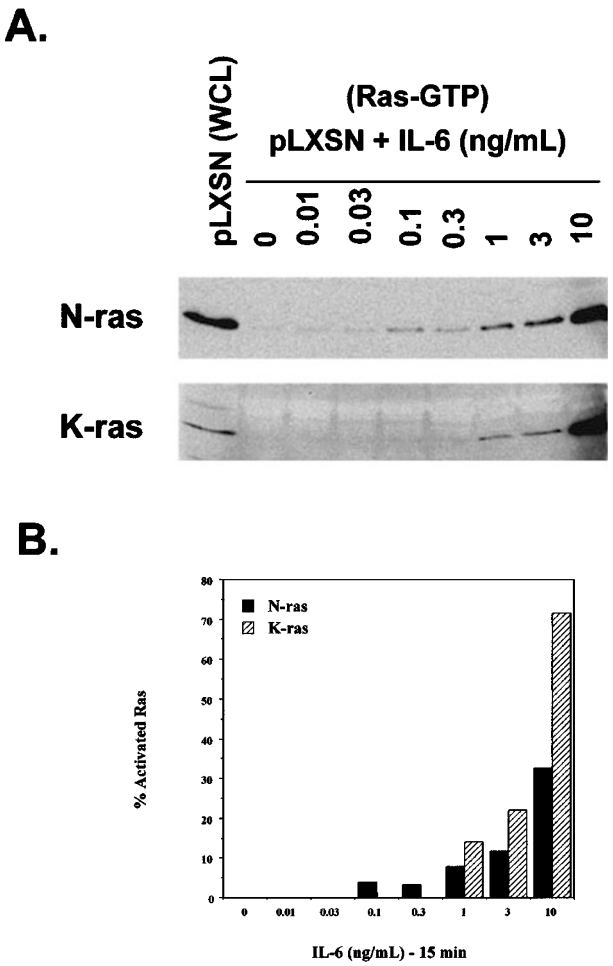

Figure 3 Activation of N- or K-ras increases with the addition of increasing amounts of interleukin-6. (**a**) ANBL6.pLXSN cells were stimulated with increasing concentrations of IL-6 for 15 min. Cells were harvested and extracts were made to measure Ras activation by ARIA using 400 μg of extract. 40 μg of whole cell lysates were used in lane 1 to measure total Ras expression. (**b**) The percentage of activated ras was measured by comparing the density of the N- or K-ras-GTP band with the total N- or K-ras expression. The percentage was adjusted to account for the 10-fold higher protein concentration used for ARIA as well as the 35 or 39% efficiency of ARIA for N- or K-ras activation respectively

the endogenous N- or K-ras in these cell lines as indicated by the presence of the faster migrating bands.

The activated Ras interaction assay is semi-quantitative

In order to quantify Ras activation induced by IL-6 we first needed to show that the assay is titratable. Cell extracts were prepared from ANBL6.Nras12.EE or ANBL6.Kras12.EE cell lines. Since these cell lines have either a constitutively active N- or K-ras that is tagged and can be distinguished from the endogenous N- or K-ras, they allow us to calculate the efficiency of the ARIA assay. The level of total tagged Ras and activated tagged Ras is the same since all of it is constitutively GTP-bound. The protein concentration was determined for each extract and varying amounts were used for the ARIA. The N- or K-ras-GTP that was pulled down was used for Western blot analysis and compared to N- or K-ras in whole cell lysates (Figure 1c). Increasing amounts of both N- and K-ras-GTP were detected when the amount of cell extract was increased. The amount of Ras-GTP recovered from 50 μg of extract used in the ARIA was slower than in the same amount of whole cell extracts. This is probably due to inefficient binding of Ras-GTP to the GST-RBD as well as loss of some protein during the subsequent wash steps. Using densitometry, and comparing the signal from the ARIA samples to the

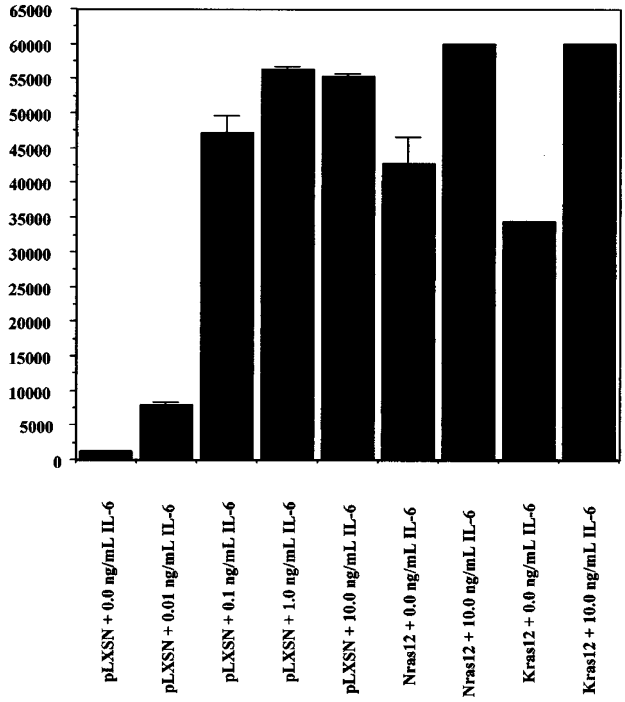

Figure 4 Proliferation of ANBL6 cells correlates with Ras activation. Cells were incubated with or without varying concentrations of IL-6 for 2 days and then pulsed with [^{3}H] thymidine. [^{3}H] Thymidine incorporation was measured by liquid scintillation counting. Values represent the mean of duplicate samples

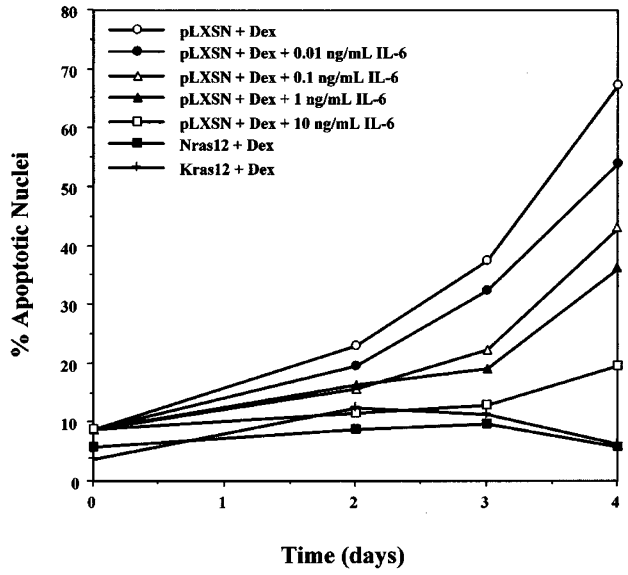

Figure 5 Protection from dexamethasone-induced apoptosis correlates with activation of N- or K-ras. ANBL6.pLXSN incubated with increasing levels of IL-6, ANBL6.Nras12.EE, and ANBL6.Kras.EE cells were treated with 1 μM dexamethasone. Cells were harvested at the indicated time point for apoptosis analysis measured by propidium iodide staining

expected signal based on the amount of Ras in whole cell lysates, the efficiency of the assay was calculated to be about 35% for N-ras and 39% for K-ras.

Interleukin-6 induces transient activation of both N- and K-ras that is increased with increasing levels of interleukin 6

Since IL-6 is able to activate both N- and K-ras, we wanted to determine what percentage of the total Ras pool is activated. ANBL6.pLXSN cells were treated with 1 ng/ml IL-6 and harvested at the indicated time points to make cell extracts. Equal amounts of protein (400 μg) were used from each time point for the ARIA assay. Both N- and K-ras were activated by 5 min in response to IL-6 (Figure 2a). Activated N- and K-ras were both detectable at 40 min, but lost by 60 min after treatment with IL-6. Densitometry and calculations based on the whole cell extract control and 35 to 39% efficiency for ARIA show that Ras activation peaked at about 11 to 18% of the total N- or K-ras pool respectively (Figure 2b).

To determine whether increasing concentrations of IL-6 lead to increasing levels of Ras activation, ANBL6.pLXSN cells were treated with varying amounts of IL-6 for 15 min. Extracts were made and equal amounts of protein were used for the ARIA assay. Figure 3 shows that activation of both N- and K-ras induced by IL-6 is titratable. Significant levels of N-ras-GTP were detectable with 0.1 ng/ml of IL-6. K-ras-GTP was detectable when cells were treated with 1 ng/ml of IL-6. K-ras activation was as high as 70%

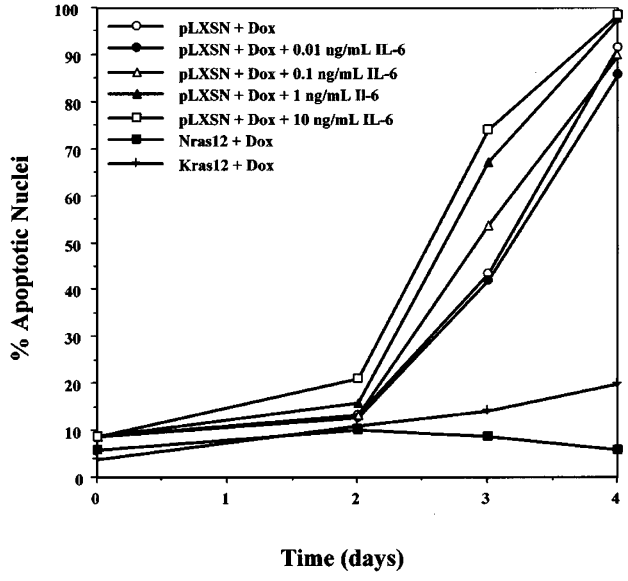

Figure 6 Interleukin-6 and Ras activation enhance doxorubicin-induced apoptosis in ANBL6.pLXSN cells, but the ANBL6.Nras12.EE and ANBL6.Kras12.EE cells are protected. ANBL6.pLXSN incubated with increasing levels of IL-6, ANBL6.Nras12.EE, and ANBL6.Kras.EE cells were treated with 250 nM doxorubicin. Cells were harvested at the indicated time point for apoptosis analysis measured by propidium iodide staining

of the total K-ras pool when stimulated with 10 ng/ml of IL-6. The decreased threshold for K-ras activation is most likely due to the sensitivity of the antibody used for the Western blot rather than an actual decrease in the activation threshold because the signal from the K-ras antibody is lower than the signal from the N-ras antibody and because once K-ras activation is detectable, the percentage of activated K-ras is similar to N-ras.

Activation of Ras by interleukin-6 correlates with proliferation

Both IL-6 and activating Ras mutations have been shown to induce proliferation of myeloma cells. We wanted to determine whether activation of N- or K-ras induced by IL-6 correlated with proliferation. ANBL6.pLXSN, ANBL6.Nras12.EE, or ANBL6.Kras12.EE cells were treated with different concentrations of IL-6 and proliferation was measured by incorporation of [^{3}H]thymidine. The ANBL6.pLXSN cell line has little [^{3}H] incorporation in the absence of IL-6, but increases in a dose-dependent manner with the addition of IL-6 (Figure 4). Proliferation plateaus above 1 ng/ml IL-6. Introduction of either the N-ras12 or K-ras12 gene induces proliferation even in the absence of IL-6 and is only slightly enhanced by additional IL-6.

Activation of Ras by interleukin-6 correlates with protection from dexamethasone-induced, but not doxorubicin-induced apoptosis

Since it has been previously shown that both IL-6 and activating Ras mutations are able to protect myeloma cells from dexamethasone induced apoptosis (Billadeau *et al*., 1997; Chauhan *et al*., 1997a; Hardin *et al*., 1994; Rowley *et al*., 2000), we wanted to determine whether the concentration of IL-6 and subsequent Ras activation correlated with the level of protection. ANBL6.pLXSN cells were incubated with increasing levels of IL-6 and then treated with dexamethasone. Higher concentrations of IL-6 lead to better protection from apoptosis (Figure 5). At 10 ng/ml IL-6, the apoptotic protection is almost at the same level as the cell lines expressing either the mutant N-*ras12* or K-*ras12* gene. This demonstrates that increasing levels of Ras activation correlates with better protection from dexamethasone-induced apoptosis.

We had previously shown that IL-6 could enhance apoptosis induced by dox (Rowley *et al*., 2000). Based on this, we wanted to determine whether increasing concentrations of IL-6 had an effect on doxorubicin-induced apoptosis. ANBL6.pLXSN cells were incubated with increasing levels of IL-6 followed by treatment with doxorubicin. In contrast to dexamethasone, ANBL6 cells became more sensitive to doxorubicin-induced apoptosis in the presence of increasing levels of IL-6 (Figure 6). Interestingly, the ANBL6.Nras12.EE and ANBL6.Kras12.EE cell lines, which have high levels of activated Ras, were resistant to doxorubicin-induced apoptosis. This suggests that even though high concentrations of IL-6 can activate a high percentage of the Ras pool, it is not enough to protect the cells from apoptosis induced by doxorubicin.

Discussion

Activation of N-ras or K-ras either by mutation or IL-6 induction plays an important role in biology of multiple myeloma. Several studies have shown that the incidence of N-*ras* and K-*ras* mutations is as high as 47% in myeloma but that H-*ras* mutations are not detected (Corradini *et al*., 1993; Liu *et al*., 1996; Neri *et al*., 1989; Paquette *et al*., 1990; Portier *et al*., 1992). We were unable to detect expression of H-ras is any of the four myeloma cell lines we tested. This would provide an explanation for the lack of H-ras mutations associated with myeloma. If H-ras were not expressed in myeloma cells, a mutation in the H-ras gene would have no effect on the growth or apoptotic response of myeloma cells.

Activation of Ras is a major component of IL-6 signal transduction. We have shown that IL-6 can activate both N-ras and K-ras in the ANBL6 cell line. In addition, IL-6-induced activation of Ras is transient and increasing concentrations of IL-6 lead to a greater level of Ras activation. While it has long been accepted that stimulation with growth factors often leads to a transient activation of signaling molecules such as Ras, the percentage of the total cellular Ras pool that is activated by IL-6 has never been reported. We were able to calculate the percentage of both N- and K-ras that is activated by IL-6. Our results suggest that only a small fraction of the total N- or K-ras pool is activated by physiological levels of IL-6. Klein *et al*. (Klein *et al*., 1989) measured the IL-6 levels in the bone marrow of patients with fulminating myeloma and showed that the concentration of IL-6 was 833+/−945 pg/ml. At that level, we expect that only about 10−15% of the Ras pool is activated.

The level of activated Ras correlates with proliferation at low levels of GTP-Ras. However, at higher levels of Ras activation, the proliferation begins to plateau. The presence of the N-*Ras12* or K-*ras12* gene is able to induce proliferation independent of IL-6, but the level of [^{3}H] thymidine incorporation is not any higher than the ANBL6.pLXSN cells treated with 0.1 ng/ml IL-6. This demonstrates that the rate-limiting factor for proliferation of ANBL6 cells is not at the level of activated Ras available. IL-6 is known to activate other transcription factors, such as Stat-3 (Rowley *et al*., 2000).

A number of studies have demonstrated that both IL-6 and activating mutations in the ras genes are able to provide protection from dexamethasone-induced apoptosis (Billadeau *et al*., 1997; Chauhan *et al*., 1997b; Lichtenstein *et al*., 1995; Rowley *et al*., 2000). In this study, we have shown that increasing concentrations of IL-6 are able to provide better apoptotic protection in response to dexamethasone. At 10 ng/ml,

IL-6 is able to activate approximately 30% of the N-ras pool and 70% of the K-ras pool. This is enough to provide protection comparable to the cells transfected with the N-*ras12* or K-*ras12* genes. Consistent with this, patients with high levels of IL-6 expression, or patients with a Ras mutation have a worse prognosis (Bataille *et al*., 1989; Liu *et al*., 1996). This may be due in part to a decreased sensitivity to therapeutic agents.

In contrast to dexamethasone-induced apoptosis, incubation of the ANBL6.pLXSN cells with IL-6 enhanced doxorubicin-induced apoptosis. Based on this, we would have expected that the constitutively active N- or K-ras would have also enhanced apoptosis, but instead, it inhibited apoptosis. One possible explanation for this may be that N- or K-ras12 mutants provide a constitutive signal of 100% of the expressed mutant Ras, while the signal induced by IL-6 is transient and does not completely activate the pool of Ras. Constitutive, high level Ras activation could lead to a constitutive activation of anti-apoptotic factors or block apoptosis. IL-6 may only transiently activate these same factors and therefore be ineffective in long-term protection from apoptosis. Indeed, in further studies we have found that Dox induces cytc release, caspase 9 and caspase 3 activation in ANBL-6 cells; whereas the ras mutant line still releases cytc in response to dox, but does not activate caspase 9 or 3 (manuscript submitted).

Materials and methods

Cell culture

The ANBL6 cell lines were as previously described (Billadeau *et al*., 1997). All cell lines were maintained in RPMI 1640 supplemented with 10% fetal calf serum, 50 U/ml each of penicillin and streptomycin, 2 mmol/l L-glutamine (Gibco BRL, Gaithersburg, MD, USA), and 0.5 ng/ml IL-6 (R&D Systems, Minneapolis, MN, USA). Cultures were incubated in the absence of IL-6 for three days to eliminate the IL-6 effect on the cells and were then centrifuged on Ficoll-Histopaque to eliminate dead cells before each experiment.

GST-RBD preparation

DH5α cells containing the pGEX-RBD expression plasmid (Taylor and Shalloway, 1996) were grown in 200 ml LB to an A600 of 0.7–0.8. Expression of GST-RBD was induced with 1 mM isopropyl-β-D-thiogalactopyranoside (IPTG) for 2 h. Bacteria were pelleted and washed once in HBS (25 mM HEPES, pH 7.5, 150 mM NaCl). The final pellet was resuspended in 10 ml of lysis buffer (20 mM HEPES, pH 7.5, 120 mM NaCl, 10% glycerol, 2 mM EDTA, 10 μg/ml leupeptin, 20 μg/ml aprotinin). Cells were lysed with a french press (SLM Instruments) at 1000 psi. Extracts were centrifuged at 15 000 *g* to pellet debris. NP-40 to 0.5% final and 500 μl packed agarose beads (equilibrated in HBS) were added to the supernatant and rocked for 30 min at 4°C. The beads were washed 7–8 times with 0.6 ml lysis buffer plus 0.5% NP-40.

Activated Ras interaction assay

$5-10\times10^6$ cells were treated as indicated, washed once in ice-cold HBS and lysed in ice cold Mg-containing lysis buffer (MLB) (25 mM HEPES, pH 7.5, 150 mM NaCl, 1% NP-40, 0.25% sodium deoxycholate, 10% glycerol, 25 mM NaF, 10 mM $MgCl_2$, 1 mM EDTA, 1 mM sodium vanadate, 10 μg/ml leupeptin, 10 μg/ml aprotinin) for 20 min at 4°C. Debris was pelleted at 15 000 *g* for 20 min. Protein concentration of the supernatants was measured using the Bio-Rad Protein Assay Kit. Equal amounts of protein were used as indicated. Lysates were rocked with 15 μl of packed GST-RBD-agarose beads at 4°C for 30 min. Beads were washed three times in 0.6 ml MLB and resuspended in SDS-PAGE sample buffer for Western blot analysis.

Western blot analysis

Proteins were separated by SDS-PAGE, transferred to a PVDF membrane (Bio-Rad, Hercules, CA, USA) and probed with anti H-ras, anti-N-ras, or anti-K-ras antibodies (Santa Cruz, Santa Cruz, CA, USA). The blots were washed and incubated with HRP conjugated secondary antibody and developed by enhanced chemiluminescence (ECL)+Plus using the manufacturer's protocol (Amersham Pharmacia, Piscataway, NJ, USA).

[^{3}H] Thymidine incorporation assay

The assay was performed as previously described (McCloskey *et al*., 1994). Briefly, cells were diluted to 2×10^5 cells/ml and 200 μl of each culture was seeded into 96 well plates in the presence of various concentrations of IL-6 as indicated. The 96 well plates were incubated for two days before the addition of one μCi of [^{3}H] thymidine (5.0 Ci/mmole; Amersham, Arlington Heights, IL, USA) to each well. The cells were further incubated for 16 h and then harvested onto glass filter paper (Skatron, Sterling, VA, USA). The [^{3}H] thymidine incorporation was measured by liquid scintillation counting (Beckman, Arlington Heights, IL, USA).

Apoptosis measurement

Cells were washed in RPMI and diluted to 2×10^5 cells/ml with RPMI 1640 with 1 mM doxorubicin or 250 nM doxorubicin or IL-6 as indicated. Cells were harvested and pelleted at the indicated time points. Cell pellets were resuspended with 0.5 ml hypotonic lysis solution (50 μg/ml propidium iodide (Sigma, St. Louis, MO, USA), 0.1% sodium citrate, and 0.1% triton X-100) and incubated at 4°C for at least 3 h in the dark. Nuclear staining was analysed on a FACSCalibur flow cytometer (Becton Dickinson, San Jose, CA, USA).

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